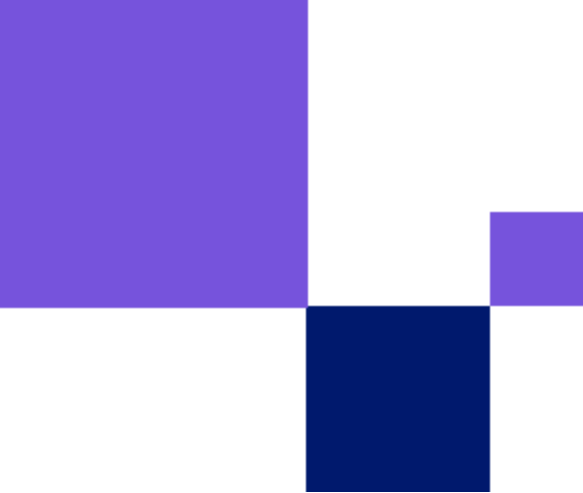


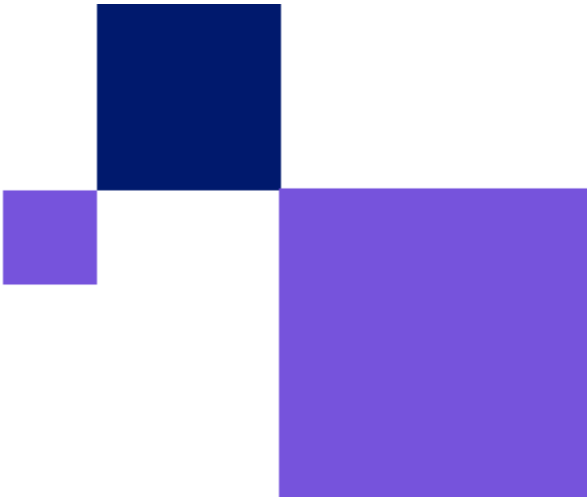


Foundations of Financial Accounting (ACC107)

Lecture 11

Chapter 29

(The Appraisal of Company Financial Statements
using Ratio Analysis)



Learning outcomes

By the end of this lecture and with your own self study you should be able to:

- Explain the purposes of ratio analysis;
- Compute, explain and evaluate various measures of a company's profitability;
- Compute, explain and evaluate various ratios used in the appraisal of working capital control;
- Compute, explain and evaluate various measures of return on investment;
- Compute, explain and evaluate various measures of liquidity and solvency (gearing/leverage);
- Discuss the limitations of ratio analysis.

Illustration (1)- Accounting Ratios to Evaluate Performance/Profitability/Return(1/2)

1. Comparisons between entities – which of the following companies is the most profitable?

	<i>Co. A</i>	<i>Co. B</i>
	£'000	£'000
Profit	200	1,000
Net assets	500	10,000

Illustration (1)- Accounting Ratios to Evaluate Performance/Profitability/Return(2/2)

1. Comparisons between entities – which of the following companies is the most profitable?

	<i>Co. A</i>	<i>Co. B</i>
	£'000	£'000
Profit	200	1,000
Net assets	500	10,000
Return on assets invested	40%	10%

Illustration (2)- Accounting Ratios to Evaluate Performance/Profitability/Return(1/2)

2. Comparisons over time – has there been an increase or decrease in profitability from one year to the next?

	Previous Year	Current Year
	£'000	£'000
Profit	900	1,000
Net assets	8,000	10,000

Illustration (2)- Accounting Ratios to Evaluate Performance/Profitability/Return(2/2)

2. Comparisons over time – has there been an increase or decrease in profitability from one year to the next?

	Previous Year	Current Year
	£'000	£'000
Profit	900	1,000
Net assets	8,000	10,000
Return	11.25%	10%

- In both comparison between entities and comparison over time;
 - a **comparison of the absolute profits would give a misleading evaluation** which needs to be made in terms of the return on assets invested.

The Functions of Ratio Analysis

The main function of ratio analysis is to enable users of published financial statements to evaluate the financial performance and financial position of the reporting entity for the purpose of share trading, lending and/or voting decisions. This usually takes the form of:

1. Comparisons with other entities (inter-firm)

- e.g. Compare Company A with B in the same industry; **and/or**

2. Comparisons over time (time series analysis).

- e.g. Compare Company A's Profits between 2020 and 2021

The Functions of Ratio Analysis

Inter-firm comparison

APPLE INC					WALMART INC					COSTCO WHOLESALE CORP				
AAPLO 148.11 USD (-1.96%) MKT CAP: 2,356.15 B					WMT 153.07 USD (0.43%) MKT CAP: 415.47 B					COSTO 533.66 USD (-0.16%) MKT CAP: 236.2 B				
▼ FUNDAMENTAL RATIOS					▼ FUNDAMENTAL RATIOS					▼ FUNDAMENTAL RATIOS				
Ratio	AAPL (Sep-22)	Growth		Industry	Ratio	WMT (Jan-22)	Growth		Industry	Ratio	COST (Aug-22)	Growth		Industry
Operating Margin	27.6%	(0.9%) ▼	4Q	8.3%	Operating Margin	4.4%	5.7% ▲	4Q	3.9%	Operating Margin	3.5%	(0.1%) ▼	4Q	3.9%
Net Margin	23.0%	(1.7%) ▼	4Q	6.5%	Net Margin	2.4%	3.7% ▲	4Q	2.3%	Net Margin	2.6%	(0.1%) ▼	4Q	2.3%
Asset Turnover	0.26	6.9% ▲	4Q	0.24	Asset Turnover	0.62	3.3% ▲	4Q	0.60	Asset Turnover	1.13	4.7% ▲	4Q	0.60
ROE	38.1%	5.8% ▲	4Q	1.6%	ROE	4.3%	6.9% ▲	4Q	4.2%	ROE	9.2%	(0.6%) ▼	4Q	4.2%
Debt/Equity	2.37	19.1% ▲	4Q	0.07	Debt/Equity	0.51	(14.8%) ▼	4Q	0.68	Debt/Equity	0.32	(25.5%) ▼	4Q	0.68

Time series analysis



The Classification of Ratios

1. Measures of **profitability**
2. Measures of the **control of working capital** (operating efficiency)
3. Measures of **return on investment**
4. Measures of **liquidity**
5. Measures of **gearing/leverage**

Measures of Company Profitability

- Return on capital employed (ROCE)
- Profit margin
- Asset turnover ratio

Return on Capital Employed (ROCE)

- ROCE is a measure of **profitability** that indicates how **efficiently and effectively** a company has utilised its assets/capital during a given accounting period.
- However, the use of historical cost data in times of rising prices overstates the 'real' return, and encourages the retention of old assets.

Return on Capital Employed (ROCE)

$$\frac{\text{Profit before tax, interest and dividends}}{\text{Net capital employed}} \times 100$$

Where,

net capital employed
= shareholders interests + non-current liabilities
= total assets – current liabilities.

Net capital employed at the start or end of the year, or an average for the year?

Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
	£'000	£'000
Revenue	5,280	4,900
Cost of sales	(3,090)	(3,000)
Gross profit	2,190	1,900
Distribution costs	(560)	(500)
Administrative expenses	(230)	(450)
Finance costs	(400)	(600)
Profit before taxation	1,000	350
Income tax expense	(250)	(100)
Profit after taxation	750	250
<i>Statement of financial position</i>		
Non-current assets at cost	9,470	8,000
Aggregate depreciation	(2,860)	(2,400)
	6,610	5,600
Current assets		
Inventories	550	400
Trade receivables	1,070	900
Bank and cash	1,130	2,000
	2,750	3,300
Total assets	9,360	8,900
EQUITY AND LIABILITIES		
Equity		
Called-up share capital: 2,000,000 equity shares of £1 each	2,000	2,000
Retained earnings	2,380	1,630
Total equity	4,380	3,630
Non-current liabilities		
10% loan stock of £100 each	4,000	5,000
Total non-current liabilities	4,000	5,000
Current liabilities		
Trade payables	730	220
Current tax payable	250	50
Total current liabilities	980	270
Total liabilities	4,980	5,270
Total equity and liabilities	9,360	8,900

Profit before tax, interest and dividends
Net capital employed

$$\frac{£1,000,000 + £400,000}{£4,380,000 + £4,000,000} \times 100 = 16.7 \text{ per cent}$$

Alternatively,

$$\frac{£1,000,000 + £400,000}{£9,360,000 - £980,000} \times 100 = 16.7 \text{ per cent}$$

Compared to 20X8 - 11.0 per cent

Worked example 29.1

$$\text{ROCE} = \text{Profit margin} \times \text{Asset-turnover ratio}$$

$$\frac{\text{Profit before tax, interest and dividends}}{\text{Net capital employed}} \times 100$$

$$\frac{\text{Profit before tax, interest and dividends}}{\text{Revenue}} \times 100$$

$$\frac{\text{Revenue}}{\text{Net capital employed}}$$

Profit Margin

$$\frac{\text{Profit before tax, interest and dividends}}{\text{Revenue}} \times 100$$

This shows what percentage of sales revenue is profit.

Different products/ industries and different sized businesses have different profit margins.

Inter-firm comparisons should therefore be limited to companies in the same industry of a comparable size.

Variations in the profit margin over time may be due to changes in sales mix, selling prices and/or unit costs.

Worked example 29.1

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Total liabilities	4,980	5,270
Total equity and liabilities	9,360	8,900

$$\frac{\text{Profit before tax, interest and dividends}}{\text{Revenue}} \times 100$$

$$\frac{£1,000,000 + £400,000}{£5,280,000} \times 100 = 26.5 \text{ per cent}$$

Compared to 20X8 – 19.4 per cent

Asset Turnover Ratio

$$\frac{\text{Revenue}}{\text{Net capital employed}}$$

This shows the amount of sales revenue generated per £ of capital employed. It is a measure of the **level of activity and productivity**.

Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
	£'000	£'000
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Total current liabilities	980	270
Total liabilities	4,980	5,270
Total equity and liabilities	9,360	8,900

Revenue
Net capital employed

$$\frac{£5,280,000}{£4,380,000 + £4,000,000} = 0.63 \text{ times}$$

Alternatively,

$$\frac{£5,280,000}{£9,360,000 - £980,000} = 0.63 \text{ times}$$

Compared to 20X8 – 0.57 times

Asset Turnover Ratio

- **Labour intensive industries usually have a high asset turnover ratio, whereas capital intensive industries normally have a low asset turnover ratio.**
 - *Why? See the denominator in the formula.*
 - *Capital intensive industries usually have a Lower 'Net Capital employed' or Higher NCE?*
 - *Inter-firm comparisons should therefore be limited to companies in the same industry.*
- Changes in the asset turnover ratio over time may be due to producing at under capacity, labour inefficiency, overstocking, etc.

APPLE INC 					WALMART INC 				
AAPL 148.11 USD (-1.96%) MKT CAP: 2,356.15 B					WMT 153.07 USD (0.43%) MKT CAP: 415.47 B				
▼ FUNDAMENTAL RATIOS					▼ FUNDAMENTAL RATIOS				
Ratio	AAPL (Sep-22)	Growth		Industry	Ratio	WMT (Jan-22)	Growth		Industry
Operating Margin	27.6%	(0.9%) ▼	4Q	8.3%	Operating Margin	4.4%	5.7% ▲	4Q	3.9%
Net Margin	23.0%	(1.7%) ▼	4Q	6.5%	Net Margin	2.4%	3.7% ▲	4Q	2.3%
Asset Turnover	0.26	6.9% ▲	4Q	0.24	Asset Turnover	0.62	3.3% ▲	4Q	0.60
ROE	38.1%	5.8% ▲	4Q	1.6%	ROE	4.3%	6.9% ▲	4Q	4.2%
Debt/Equity	2.37	19.1% ▲	4Q	0.07	Debt/Equity	0.51	(14.8%) ▼	4Q	0.68

Asset turnover and profit margin

- There is an **important relationship** between the asset-turnover ratio and the profit margin
 - ✓ **ROCE = Profit margin x Asset Turnover**
- In order to achieve a satisfactory return on capital employed (ROCE)
 - ✓ A company with a low asset-turnover ratio (e.g. **capital-intensive**) will **need a high profit margin** on its products.
 - ✓ Conversely, a company with a high asset-turnover ratio (e.g. **labour-intensive**) will **only require a low profit margin** on its products in order to achieve a satisfactory return on capital employed

APPLE INC					WALMART INC				
AAPL 148.11 USD (-1.96%) MKT CAP: 2,356.15 B					WMT 153.07 USD (0.43%) MKT CAP: 415.47 B				
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Debt/Equity	2.37	19.1% ▲	4Q	0.07	Debt/Equity	0.51	(14.8%) ▼	4Q	0.68

In-class MCQs

Q1. If a company has a low Asset Turnover ratio, which of the following is most likely true?

- A) The company is highly efficient in generating revenue from its assets.**
- B) The company is under-utilizing its assets.**
- C) The company is generating high ROCE.**
- D) The company has a high profit margin.**

In-class MCQs

Q2. Which of the following statements is true about the relationship between Asset Turnover and Profit Margin?

- A) A higher Asset Turnover always results in a higher Profit Margin.**
- B) A higher Asset Turnover may result in a lower Profit Margin.**
- C) Asset Turnover and Profit Margin are unrelated.**
- D) A higher Profit Margin always results in a higher Asset Turnover.**

In-class MCQs

Q3. Which of the following best describes the relationship between Profit Margin, Asset Turnover, and ROCE?

- A) An increase in Profit Margin always increases ROCE.**
- B) An increase in Asset Turnover always decreases ROCE.**
- C) ROCE is determined by both Profit Margin and Asset Turnover, and improving either can increase ROCE.**
- D) Profit Margin and Asset Turnover have no impact on ROCE.**

The Appraisal of Working Capital

- Working capital refers to the level of current assets and current liabilities.
- The appraisal of working capital reveals **whether too much or too little is invested in**, for example, trade receivables and inventories relative to a company's level of activity (i.e. turnover).
- This involves identifying changes in the relative level of trade receivables and inventories which are taken as prima facie indicators of the effectiveness of credit control and inventory control respectively.

Measures of the Control of Working Capital

- Trade receivables ratio or average period of credit taken by trade customers
- Trade payables ratio or average period of credit received from trade suppliers
- Inventory turnover ratio

Trade receivables Ratio

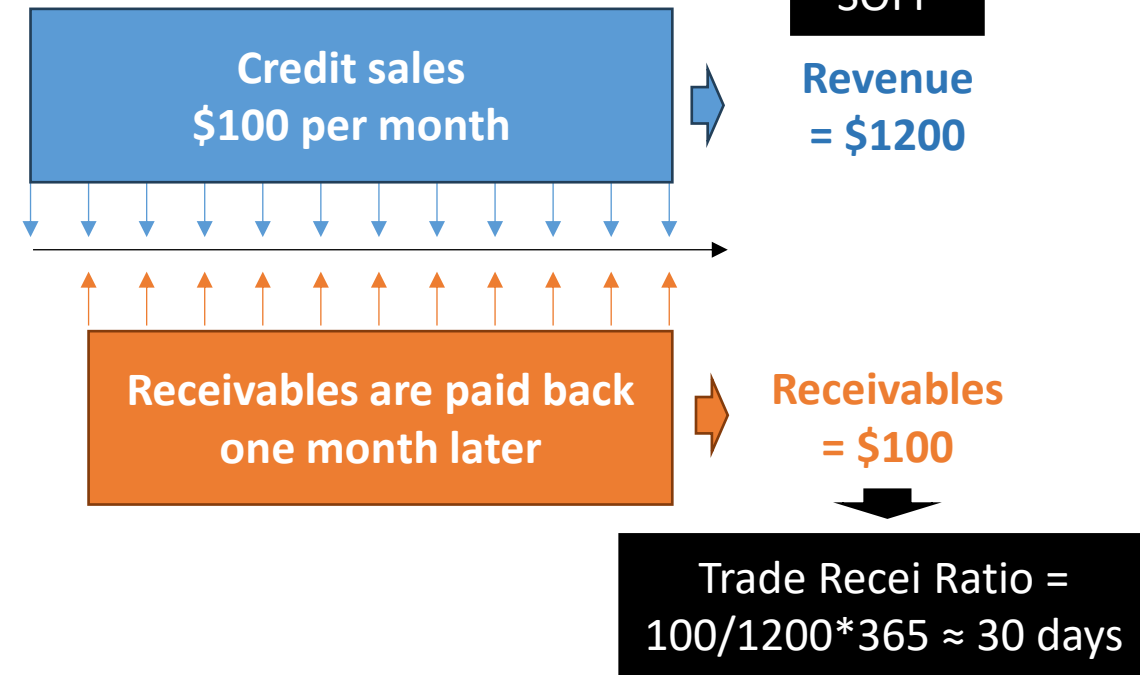
$$\frac{\text{Trade receivables}}{\text{Revenue}} \times 365$$

Normal average of around 45 – 75 days.

Changes in the proportion of cash to credit sales revenue can distort the ratio over time.

Where the period of credit is high compared with other businesses (or the average for the industry), and/or increasing over time, this indicates an inadequate credit control procedures.

- *Lower TR ratios => it take less time to convert T/R to cash.=> Good or bad for the business?*



Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
	£'000	£'000
Revenue	5,280	4,900
Cost of sales	(3,090)	(3,000)
<i>Gross profit</i>	2,190	1,900
Distribution costs	(560)	(500)
Administrative expenses	(230)	(450)
Finance costs	(400)	(600)
Profit before taxation	1,000	350
Income tax expense	(250)	(100)
<i>Profit after taxation</i>	750	250
<i>Statement of financial position</i>		
<i>Non-current assets at cost</i>	9,470	8,000
Aggregate depreciation	(2,860)	(2,400)
	6,610	5,600
<i>Current assets</i>		
Inventories	550	400
Trade receivables	1,070	900
Bank and cash	1,130	2,000
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<i>Total assets</i>	9,360	8,900
EQUITY AND LIABILITIES		
<i>Equity</i>		
Called-up share capital: 2,000,000 equity shares of £1 each	2,000	2,000
Retained earnings	2,380	1,630
<i>Total equity</i>	4,380	3,630
<i>Non-current liabilities</i>		
10% loan stock of £100 each	4,000	5,000
<i>Total non-current liabilities</i>	4,000	5,000
<i>Current liabilities</i>		
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Current tax payable	250	50
<i>Total current liabilities</i>	980	270
<i>Total liabilities</i>	4,980	5,270
<i>Total equity and liabilities</i>	9,360	8,900

$$\frac{\text{Trade receivables}}{\text{Revenue}} \times 365$$

$$\frac{\text{£1,070,000}}{\text{£5,280,000}} \times 365 = 74 \text{ days}$$

Compared to 20X8 – 67 days

Trade Payables Ratio

$$\frac{\text{Trade payables}}{\text{Cost of sales}} \times 365$$

Where the period of credit is high compared with other businesses (or the average for the industry), and/or increasing over time, this indicates that the company is delaying paying to its credit suppliers.

It may adversely affect a company's credit rating, suppliers may refuse to supply further goods on credit, and it could suggest financial weakness.

- *Lower TP ratios => The business pays T/P more quickly.*
=> Good or Bad for the business?

Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
	£'000	£'000
Revenue	5,280	4,900
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Current tax payable	250	50
Total current liabilities	980	270
Total liabilities	4,980	5,270
Total equity and liabilities	9,360	8,900

$$\frac{\text{Trade payables}}{\text{Cost of sales}} \times 365$$

$$\frac{£730,000}{£3,090,000} \times 365 = 86 \text{ days}$$

Compared to 20X8 – 26 days

Inventory Turnover Ratio

$$\frac{\text{Cost of sales}}{\text{Inventory of finished goods}}$$

This shows the number of times that a business 'turns over'/sells its average/ normal level of inventory during the accounting year.

Where the inventory turnover ratio is low compared with other businesses (or the average for the industry), and/or decreasing over time, this indicates an inadequate inventory control.

Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
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Cost of sales
Inventory of finished goods

$$\frac{£3,090,000}{£550,000} = 5.6 \text{ times}$$

Compared to 20X8 - 7.5 times

Inventory Days (not examined)

$$\frac{\text{Average Inventory}}{\text{Cost of Sale}} \times 365$$

- This ratio is also useful to understand the business's operation (although this is not introduced in the textbook).
- “Inventory days” can be simply interpreted as an average time to realise inventory into trade receivables.
 - See trade receivable ratio and trade payable ratio.

Worked example 29.1 (not examined)

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

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$$\frac{\text{Average Inventory}}{\text{Cost of Sale}} \times 365$$

$$\frac{1/2 * (£550,000 + £400,000)}{£3,090,000} \times 365 = 56 \text{ days}$$

Measures of the Control of Working Capital (Real-world examples)

AAPL.O **APPLE INC** 148.110000 c USD -2.960000 -1.9594 % Vol

United States | NASDAQ Global Select Consolidated | Phones & Handheld Devices

Overview News & Research Price & Charts Estimates **Financials** ESG Event Ownership Debt & Credit Peers & Valuation

RATIOS - KEY METRICS Annual Standardised in Millions of U.S. Dollars

Period Annual View Standardised

	Industry Median	2022	2021	2020	2019	2018
Operating						
Avg. A/P Days	86.1	96.8	82.9	95.0	114.9	111.3
Avg. A/R Days	56.1	51.9	44.3	55.2	66.3	58.0
Inv Turnover	5.6	38.8	40.0	41.5	40.1	37.2
Avg. Inventory Days	65.2	9.4	9.1	8.8	9.1	9.8

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RATIOS - KEY METRICS Annual Standardised in Millions of U.S. Dollars

Period Annual View Standardised

	Industry Median	2022	2021	2020	2019	2018
Operating						
Avg. A/P Days	31.5	44.5	41.8	43.6	44.2	42.9
Avg. A/R Days	13.0	4.7	4.2	4.4	4.2	4.2
Inv Turnover	11.6	8.5	9.4	8.9	8.8	8.6
Avg. Inventory Days	31.5	43.3	38.9	41.1	41.8	42.6

In-class MCQs

Q4. What is a high Inventory Turnover Ratio generally indicative of?

- A) The company is not selling its inventory quickly.**
- B) The company is efficiently converting its inventory into sales.**
- C) The company has high holding costs.**
- D) The company has too much inventory.**

In-class MCQs

Q5. If a company's Trade Payable Days ratio is increasing, this suggests:

- A) The company is paying its suppliers faster.**
- B) The company is delaying payments to its suppliers.**
- C) The company is reducing its inventory.**
- D) The company is increasing its accounts receivable collection period.**

Measures of Investment Return

- Dividend yield
- Dividend cover ratio
- Earnings per share (EPS)
- Price-earnings (P-E) ratio
- Return on equity (ROE)

Dividend Yield

$$\frac{\text{Annual equity dividend per share}}{\text{Market value of equity shares}} \times 100$$

This measures the equity shareholder's annual return on investment, and may be compared with what could be obtained by investing in some other company.

However, such comparisons may be misleading for two reasons.

- Companies have different risk characteristics.
- Companies have different dividend policies.

Worked example 29.1

The following is an extract from the published financial statements of A. Harry plc, for the year ending 31 January 20X9:

<i>Statement of profit or loss</i>		
	20X9	20X8
	£'000	£'000
Revenue	5,280	4,900
Cost of sales	(3,090)	(3,000)
Gross profit	2,190	1,900
Distribution costs	(560)	(500)
Administrative expenses	(230)	(450)
Finance costs	(400)	(600)
Profit before taxation	1,000	350
Income tax expense	(250)	(100)
Profit after taxation	750	250
<i>Statement of financial position</i>		
Non-current assets at cost	9,470	8,000
Aggregate depreciation	(2,860)	(2,400)
	6,610	5,600
Current assets		
Inventories	550	400
Trade receivables	1,070	900
Bank and cash	1,130	2,000
	2,750	3,300
Total assets	9,360	8,900
EQUITY AND LIABILITIES		
Equity		
Called-up share capital: 2,000,000 equity shares of £1 each	2,000	2,000
Retained earnings	2,380	1,630
Total equity	4,380	3,630
Non-current liabilities		
10% loan stock of £100 each	4,000	5,000
Total non-current liabilities	4,000	5,000
Current liabilities		
Trade payables	730	220
Current tax payable	250	50
Total current liabilities	980	270
Total liabilities	4,980	5,270
Total equity and liabilities	9,360	8,900

Additional information

1. The equity shares and loan stock are currently quoted on the London Stock Exchange at £4 and £90, respectively.

The figures for 20X8 were £2.40 and £85.

2. A final dividend for the year of £400,000 is proposed (no interim dividend was paid in the year).

The dividend for the year 20X8 was £250,000.

Annual equity dividend per share x 100
Market value of equity shares

$$\frac{\text{£400,000} / 2,000,000}{\text{£4}} \times 100 = 5 \text{ per cent}$$

Compared to 20X8 – 5.2 per cent

Dividend Cover Ratio

$$\frac{\text{Profit after taxation}}{\text{Annual equity dividend}}$$

This indicates how likely it is that the company will be able to maintain future dividends at their current level if profits were to fall in future years.

It is thus a measure of risk.

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The dividend for the year 20X8 was £250,000.

Profit after taxation Annual equity dividend

$$\frac{£750,000}{£400,000} = 1.875$$

Compared to 20X8 – 1

Earnings Per Share (EPS)

$$\frac{\text{Profit after taxation}}{\text{Number of equity shares}}$$

The EPS is a **measure of financial performance**, where performance is defined from the equity shareholders point of view as relating to the profit available for distribution as equity dividends.

The trend in EPS over time indicates growth or otherwise in the profit attributable to each equity share. Inter-firm comparisons of EPS are not advisable.

Worked example 29.1

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2. A final dividend for the year of £400,000 is proposed (no interim dividend was paid in the year).

The dividend for the year 20X8 was £250,000.

Profit after taxation
Number of equity shares

$$\frac{£750,000}{2,000,000} = £0.375$$

Compared to 20X8 - £0.125

Price Earnings (P-E) Ratio

$$\frac{\text{Market price of each equity share}}{\text{Earnings per share}}$$

The P-E ratio represents the number of years that the investors can recover their investments at the current price and earnings level. It is essentially a payback period.

The P-E ratio is also a reflection of the **expected growth potential of a company**.

- If investors think that a company's earnings are going to decline, the P-E ratio will be lower than the norm.
- Conversely, if investors think that a company's earnings are going to rise, the P-E ratio will be higher than the norm.
- Both of these occur because of decreases and increases in the share price respectively.

Worked example 29.1

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Market price of each equity share Earnings per share

$$\frac{\text{£4}}{\text{£0.375}} = 10.67$$

Compared to 20X8 - 20

Return on Equity/Shareholders Interests (ROE)

$$\frac{\text{Profit after taxation}}{\text{Equity}} \times 100$$

This is a measure of the **equity shareholders' return on investment** which is used to evaluate profitability, where profitability is defined from the shareholders' point of view as relating to the profit available for distribution as dividends.

It is said to indicate how efficiently and effectively a company's management has utilised the shareholders' investments to generate profits.

The ratio may be used to compare the profitability of different companies and/or to examine trends over time.

Worked example 29.1

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Additional information

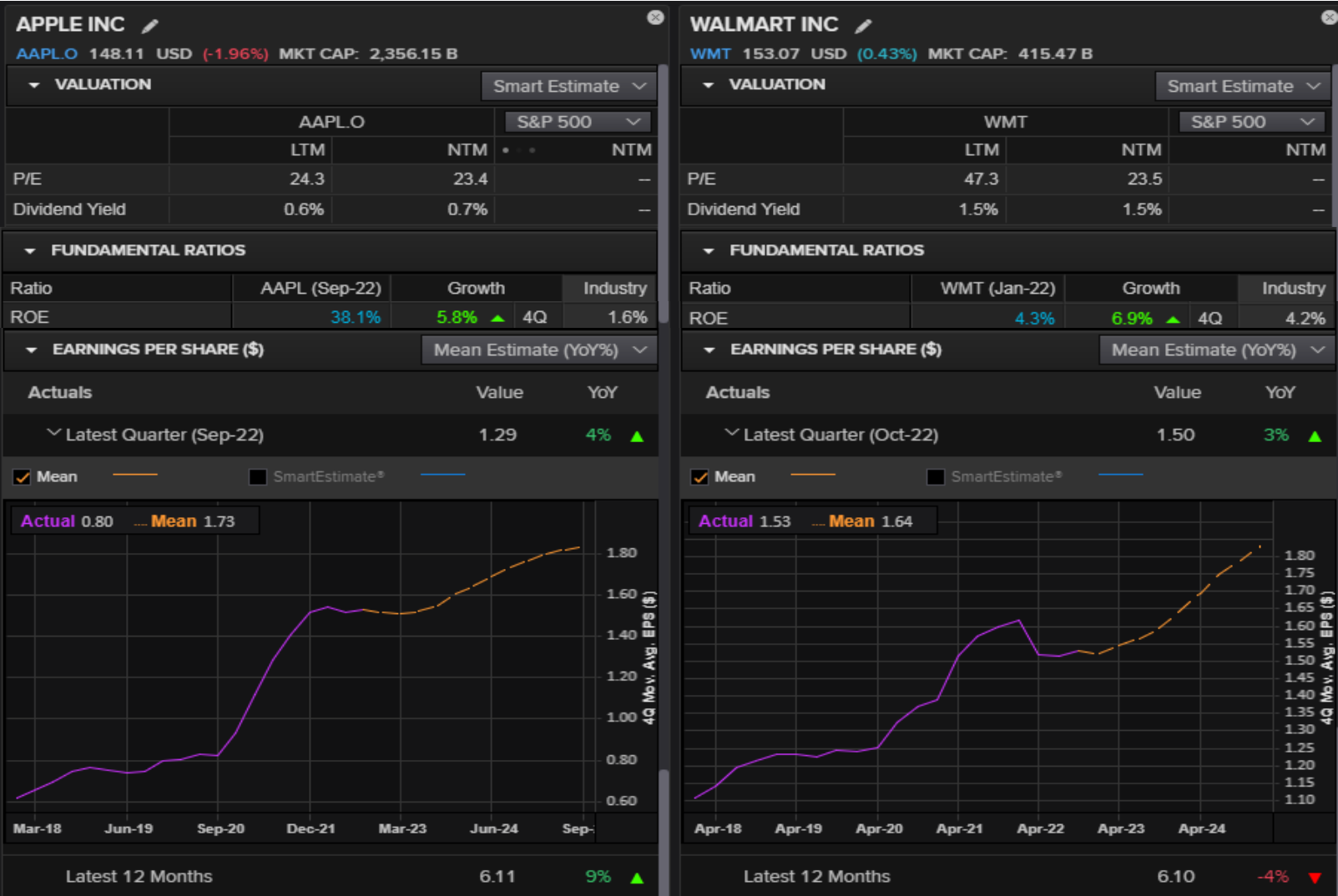
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$$\frac{\text{Profit after taxation}}{\text{Equity}} \times 100$$

$$\frac{£750,000}{£4,380,000} \times 100 = 17.1 \text{ per cent}$$

Compared to 20X8 – 6.8 per cent

Measures of Investment Return (Real-world examples)



In-class MCQs

Q6. Which of the following statements about the P/E ratio is correct?

- A) A high P/E ratio indicates that a company's stock is undervalued.**
- B) The P/E ratio is calculated using the company's dividends.**
- C) A low P/E ratio may indicate that a company's stock is undervalued or that the company has low growth prospects.**
- D) The P/E ratio is irrelevant in evaluating a company's performance.**

In-class MCQs

Q7. Which of the following is true regarding Dividend Yield?

- A) It measures the amount of profit a company generates relative to its equity.**
- B) It indicates the return on investment from dividends relative to the stock price.**
- C) It reflects how much a company is willing to pay in earnings to its shareholders.**
- D) It is an indicator of how much the company's stock price will appreciate.**

The Nature of Liquidity and Solvency

Liquidity refers to the ease with which an asset **can be turned into cash without loss**.

Current assets are shown in published statements of financial position in their reverse order of liquidity.

Solvency refers to whether or not a business is able to **pay its long-term debts** when they become due.

Sometimes even profitable businesses become insolvent, usually due to overtrading. A company that is insolvent is required by law to cease trading and become bankrupt/liquidation.

Measures of Liquidity and Solvency

Liquidity

- Working capital ratio (= Current ratio)
- Liquidity ratio (= Quick ratio or Acid-test ratio)

Solvency

- Gearing/Leverage

Working capital ratio (= Current ratio)

$$\frac{\text{Current assets}}{\text{Current liabilities}}$$

This is a measure of the extent to which current liabilities are covered by current assets.

As a generalisation, the current ratio should be between 1.5 and 2 although this depends on the type of industry.

Worked example 29.1

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<i>Total equity and liabilities</i>	<u>9,360</u>	<u>8,900</u>

Current assets
Current liabilities

$$\frac{\text{£2,750,000}}{\text{£980,000}} = 2.8$$

Compared to 20X8 – 12.2

Liquidity ratio (= Quick ratio or Acid-test ratio)

$$\frac{\text{Current assets} - \text{inventories}}{\text{Current liabilities}}$$

This ratio excludes inventories because these are not as liquid as other current assets. It is thus a better measure of liquidity than the current ratio.

As a generalisation, the liquidity ratio should be at least 1 although this depends on the type of industry (e.g. food retailers may have a ratio of less than one because they turn over their inventory rapidly).

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Current assets – inventories
Current liabilities

$$\frac{\text{£2,750,000} - \text{£550,000}}{\text{£980,000}} = 2.24$$

Compared to 20X8 – 10.74

The Nature of Gearing/Leverage

Gearing, or *leverage* as it is called in the USA, refers to the relationship between the amount of debt capital and the amount of equity capital (i.e. equity shares).

The debt capital includes loan stock, debentures, bank loans, mortgages and any other long term borrowing.

High gearing: $\text{debt capital} > \text{equity capital}$

Low gearing: $\text{debt capital} < \text{equity capital}$

The gearing ratio is a measure of the '*financial risk*' attaching to a company's equity shares which arises because of the prior claim that fixed interest capital has on the annual income and assets (in the event of liquidation).

Measures of the Gearing Ratio

Gearing ratio:

$$\frac{\text{Debt capital}}{\text{Debt capital} + \text{Equity capital}} \times 100$$

- Using nominal value (incl. reserves and retained profits)

$$\frac{\text{Nominal value of debt capital}}{(\text{Nominal value of debt capital} + \text{Nominal value of equity shares} + \text{Reserves and retained profits})} \times 100$$

** Gearing ratio can be calculated using market value as well (* Not introduced in the lecture)*

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The dividend for the year 20X8 was £250,000.

$$\frac{\text{Debt capital}}{\text{Debt capital} + \text{Equity capital}} \times 100$$

$$\frac{£4,000,000}{£4,000,000 + £4,380,000} \times 100 = 47.7 \text{ per cent}$$

Compared to 20X8 - 57.9 per cent

Measures of Solvency, Liquidity and Leverage (Real-world examples)

AAPL.O **APPLE INC** 📶 🕒 ▼ 148.110000 c USD -2.960000 -1.9594 % Vol

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⊖ Liquidity						
Quick Ratio	0.91	0.85	1.02	1.33	1.50	1.10
Current Ratio	1.62	0.88	1.07	1.36	1.54	1.13
⊖ Leverage						
Debt/Equity	0.27	2.39	1.99	1.73	1.19	1.07

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⊖ Liquidity						
Quick Ratio	0.63	0.28	0.49	0.22	0.23	0.20
Current Ratio	1.22	0.93	0.97	0.79	0.80	0.76
⊖ Leverage						
Debt/Equity	0.72	0.51	0.60	0.73	0.75	0.55

In-class MCQs

Q8. Which of the following is an ideal Current Ratio for most businesses?

A) Less than 1

B) 1

C) 1.5 to 2

D) 5 or more

In-class MCQs

Q9. If a company's Quick Ratio is less than 1, it indicates that:

- A) The company has sufficient short-term assets to pay off its current liabilities.**
- B) The company may face liquidity issues in meeting its short-term obligations.**
- C) The company has excessive inventories.**
- D) The company has excellent profitability.**

In-class MCQs

Q10. Which of the following best describes a high Leverage Ratio?

- A) The company is less dependent on debt financing and more reliant on equity financing.**
- B) The company is highly leveraged and uses a significant amount of debt to finance its operations.**
- C) The company has low risk and is more solvent.**
- D) The company is likely to have a low return on equity.**

The Limitations of Ratio Analysis

- Different accounting policies results in misleading comparisons.
- Historical cost data distorts comparisons.
- Surrogate data may have to be used.
 - e.g.) the trade receivables' (and trade payables') collection period appears to be extremely low, which may be because revenue (cost of sales) includes cash sales (purchases).
- Performance criteria/yardsticks are not appropriate for all types of industries.
- Comparisons using ratios need to consider the current economic climate, type of industry, and the company's specific circumstances, including management policies.
- A single ratio provides limited information, but a group of related ratios gives a general picture of the performance or financial position.

Homework

- ▶ Review the content of this lecture
- ▶ Attempt the tutorial questions on time

Thanks for listening

